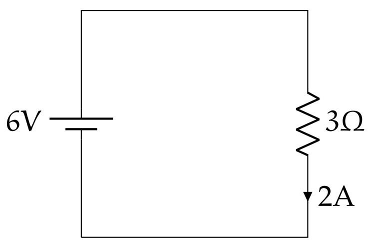
Chapter 1

DC Circuit Analysis

In the most basic circuit, you have only a battery and a resistor:

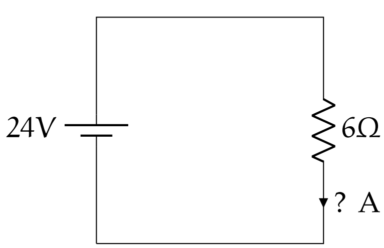


For this situation, you only need Ohm's Law: $V = IR$. In this case, $6V = 3\Omega \times 2A$.

Exercise 1 Ohm's Law

Working Space

How many amps are going around the circuit?

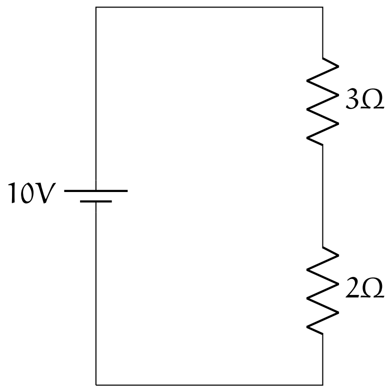


Answer on Page 9

1.1 Resistors in Series

When you have two resistors wired together in a long line, we say they are "in series." If you have two resistors R_1 and R_2 wired in series, the total resistance is $R_1 + R_2$.

In this diagram, for example, the total resistance is 5Ω.



The current flowing through the circuit, then, is $10/5 = 2A$.

By Ohm's law, the voltage drop across the upper resistor is $IR = 2A \times 3\Omega = 6V$.

The voltage drop across the lower resistor is $IR = 2A \times 2\Omega = 4V$.

1.1.1 Kirchhoff's Voltage Law

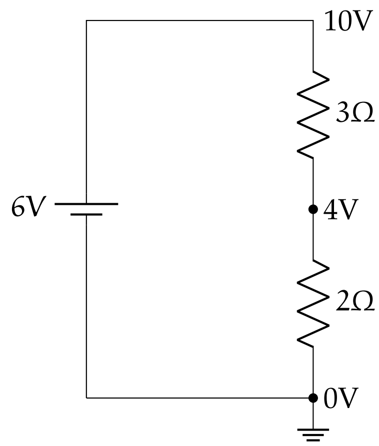
Notice that the battery pumps the voltage up to 10V, then the two resistors drop it by exactly 10V. This is known as "Kirchhoff's Voltage Law":

Kirchhoff's Voltage Law

As you make a loop around a circuit, the sum of the voltage increase must equal the sum of the voltage decrease. Mathematically, this is

$$\sum_{i=1}^{n} V_i = 0 \tag{1.1}$$

The negative end of the battery is connected to "ground" (it has zero voltage). We can then draw a diagram with the voltages (That symbol in the lower right represents a connection to ground).

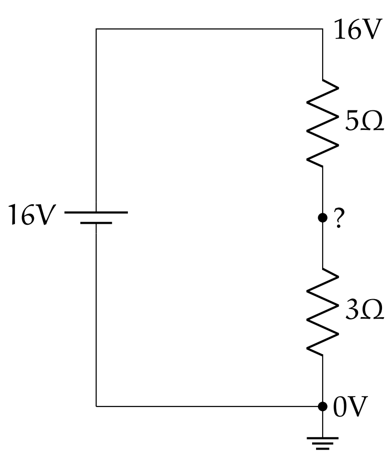


Exercise 2 Resistors In Series

Working Space

What is the current going around the circuit?

What is the voltage drop across each resistor?



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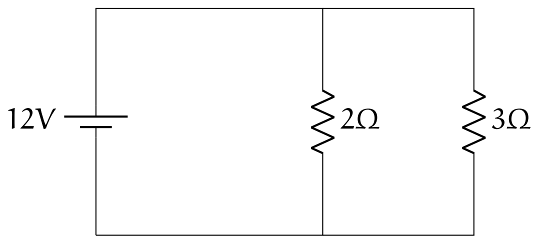
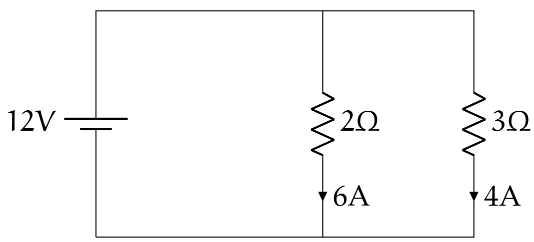
1.2 Resistors in Parallel

Observe the following circuit. Note that the current can go two different paths.



There is 12 volts pushing current through both resistors. So 6A will go through the 2Ω resistor and 4A will go through the 3Ω resistor. Note that even though there is a path of least resistance (2Ω), the current is still divided evenly among both branches.



Thus, a total of 10 A will be going through the battery.

Imagine you are a battery. You can't see that you have two resistors. What does it feel like to you? $\frac{V}{I} = R$, and $V = 12$ and $I = 10$. This means the effective resistance of the two resistors in parallel is $\frac{12}{10}$ or $\frac{6}{5}\Omega$.

Resistance in Parallel

If you have several resistances $R_1, R_2, \ldots, R_n$ wired in parallel, their effective resistance R_t is given by

$$\frac{1}{R_t} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots + \frac{1}{R_n}$$

In our example:

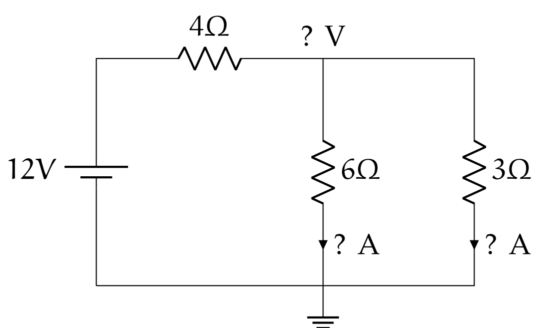
$$\frac{1}{R_t} = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$$

Thus, $R_t = \frac{6}{5}\Omega$.

Exercise 3 Resistors In Parallel

Working Space

What is the current going through the battery? What is the drop over the 4Ω resistor? What is the current in each branch?



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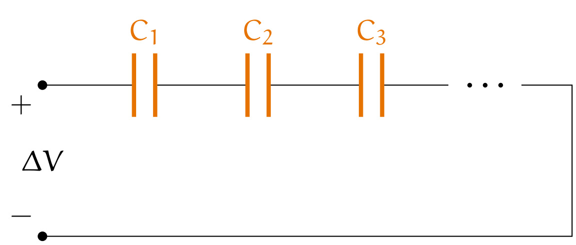
1.3 Capacitors in series and parallel

Recall that a capacitor is similar to a battery in that it stores voltage, but it is stored in the form of an electric field.

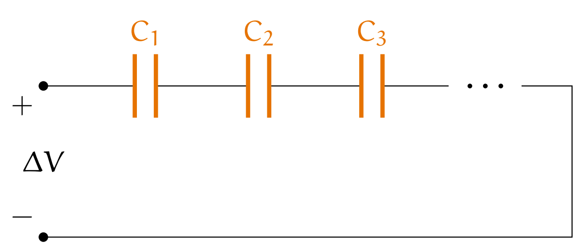


Capacitors are represented with two differently colored lines. Here is an circuit diagram of capacitors in **parallel** Capacitors are said to be in parallel if they share the same potential difference. You may see this represented mathematically:

$$C_p = \sum_i C_i \tag{1.2}$$

Capacitors are **series** if they all share the same charge magnitude. This would be mathematically represented as

$$\frac{1}{C_S} = \sum_i \frac{1}{C_i} \tag{1.3}$$



A common mistake in calculating series capacitance is neglecting to take the reciprocal to find the equivalent capacitants after finding the reciprocal of each individual capacitor. If you have n capacitors in series all with value C, the total capacitance is $C_S = \frac{C}{n}$.

1.4 Kirchhoff's Current Law

Kirchhoff's Current Law states that all currents coming into a junction (or node) must equal the currents leaving that junction.

Why? Because first of all, energy is always conserved, meaning it cannot be lost or destroyed within the equations in the first place.

Secondly, each circuit is conservative, meaning it cannot lose voltage, and thus, cannot lose current. As you go around a loop, the starting point must have the same chargable amount as it did when you began, so any increases and decreases along the loop have to cancel out for a total change of zero.

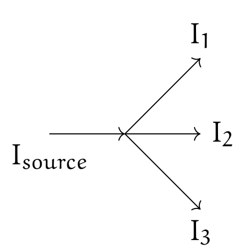
Kirchhoff's Current Law

The sum of current into a junction equals the sum of current out of the junction.

Mathematically, this can be expressed as

$$\sum_{i=1}^{n} I_i = 0 \tag{1.4}$$

where n is the total number of branches with currents flowing towards or away from the node.



Kirchhoff's Current Law states that all currents going out of a junction must equal the current going into a junction. This is aligned with the Conservation of energy: **the total energy of an isolated system remains constant**. In this case,

$$I_{source} = I_1 + I_2 + I_3 \tag{1.5}$$

Exercise 4 Kirchhoff's Current Law

At a junction, 5 conductors meet at a single common junction. Given Branch Currents:

- $i_1 = 5$ A (entering the node)
- $i_2 = 3$ A (leaving the node)
- $i_3 = 2$ A (entering the node)
- $i_4 = 9$ A (leaving the node)

Find the direction of the unknown current i_x.

Answer on Page 10

1.5 Summary

This chapter introduces the fundamentals of Direct Current (DC) circuits, focusing on the relationship between power sources and resistive elements.

- **Basic circuits**: The analysis begins with a minimal circuit configuration consisting of a voltage source (battery) and a load (resistor).
- **Voltage** (V): The electrical potential difference provided by the battery that drives the flow of charge.
- **Resistance** (R): The property of a material or component that opposes or *resists* the flow of electric current.
- **Direct Current** (DC): A type of electrical flow where the charge moves in a single, constant direction.
- **Kirchhoff's Laws**: Kirchhoff's Laws tell us that charge cannot be lost or pile up at a junction, and any supplied charged must be dropped to zero before the charge cycles around.

In a few workbooks, we will be doing a deep dive into charge, and how charge creates electric fields, electric flux, dipoles, and spherical electric potential.

This is a draft chapter from the Kontinua Project. Please see our website (https://kontinua.org/) for more details.

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 1)

$V = IR$ so $I = \frac{V}{R} = \frac{24V}{6\Omega} = 4A$.

Answer to Exercise ?? (on page 3)

There is a total resistance of 8Ω, so your 16V will push 2A of current around the circuit.

2A going through a 5Ω resistor represents a 10V drop.

2A going through a 3Ω resistor represents a 6V drop.

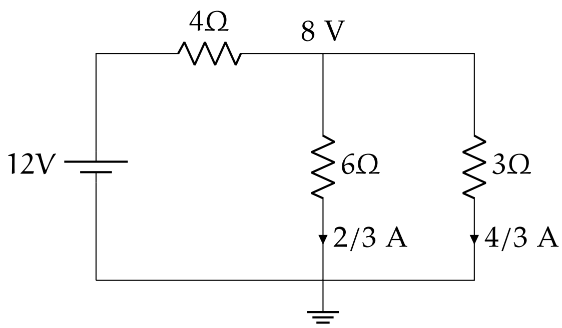
Answer to Exercise 3 (on page 5)

The effective resistance of the 6Ω and the 3Ω is 2Ω because

$$\frac{1}{R_T} = \frac{1}{6} + \frac{1}{3} == \frac{1}{2}$$

This means the battery experiences a resistance of $4\Omega + 2\Omega = 6\Omega$. A 12V will push 2A through a resistance of 6Ω.

The voltage drop across the 4Ω resistor is $2A \times 4\Omega = 8V$. Thus there will be a 4V drop across the two resistors in parallel. So 2/3 A will flow through the 6Ω resistor. 4/3 A will flow through the 3Ω resistor.



Answer to Exercise 4 (on page 7)

Let's set up the equation. Here are the entering conductors:

$$i_1, i_3$$

And leaving: i_2, i_4, i_x we are assuming i_x is leaving for the calculation.

$$i_1 + i_3 = i_2 + i_4 + i_x$$

Inputting the values gives us:

$$5 + 2 = 3 + 9 + i_x$$
$$7 = 12 + i_x$$
$$i_x = -5\,\mathrm{A}$$

The magnitude of the current is 5 A. Because the result is negative, the actual direction is *entering* the node, based on our leaving assumption.

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